TWO INSTABILITY INTERVALS FOR A SWITCHED PLANAR SYSTEM WITH ONE NON-HURWITZ MATRIX

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ABSTRACT. Let $\dot{X}_t = A_{I_t} X_t$ in $\mathbb{R}^2$, where I jumps between 0 and 1 at rate r. Malrieu asked whether A_0, A_1 can be chosen so that the set of unstable rates is a union of several intervals. For the Hurwitz matrix $A_0 = \begin{pmatrix} -3/4 & -1 \\ 1 & -3/4 \end{pmatrix}$ and the non-Hurwitz matrix $A_1 = \operatorname{diag}(5/4, -11/4)$ we prove that the top Lyapunov exponent $\lambda(r)$ is continuous, positive on $(0, 1/10]$, negative on $[4.997, 5.003]$ and positive on $[19.95, 20.05]$, so the unstable set has at least two components. The signs near $r = 5$ and $r = 20$ come from trigonometric polynomials with exact rational coefficients that solve the Poisson equation of the angular process up to an explicit error. Numerically λ changes sign twice, near $r = 1.2228$ and $r = 7.7616$. The case in which both matrices are Hurwitz, as in the example behind the question, remains open.

1. INTRODUCTION

Let A_0, A_1 be real 2×2 matrices and $r > 0$. Let $(I_t)_{t \geq 0}$ be the Markov process on $\{0, 1\}$ that jumps from 0 to 1 and from 1 to 0 at rate r, and let X solve $\dot{X}_t = A_{I_t} X_t$. Equivalently, (X, I) is the Markov process on $\mathbb{R}^2 \times \{0, 1\}$ with generator

$$Lf(x, i) = (A_i x) \cdot \nabla_x f(x, i) + r\big(f(x, 1 - i) - f(x, i)\big). \tag{1}$$

This is the setting of Section 4.1 of Malrieu's survey [10], which presents an example of Lawley, Mattingly and Reed [9]. For $\alpha > 0$ the matrices $A_0 = \left(\begin{smallmatrix} -\alpha & 1 \\ 0 & -\alpha \end{smallmatrix}\right)$ and $A_1 = \left(\begin{smallmatrix} -\alpha & 0 \\ -1 & -\alpha \end{smallmatrix}\right)$ are Hurwitz (all eigenvalues have negative real part), and so is every convex combination of them. Yet $|X_t| \to \infty$ almost surely for some α and some rates r. For this family the set of unstable rates is empty or, under Conjecture 4.3 of [10], an interval. Malrieu then asks the following.

Open question 3 of [10] (Oscillations of the Lyapunov exponent). Is it possible to choose the two 2×2 matrices A_0 and A_1 in such a way that the set of jump rates r associated to unstable processes is the union of several intervals?

The question is stated for two real 2×2 matrices. Theorem 1.1 answers it with a pair in which A_0 is Hurwitz and A_1 is not; for two Hurwitz matrices it is open (Section 6). Let Φ_t solve $\dot{\Phi}_t = A_{I_t} \Phi_t$, $\Phi_0 = \mathrm{Id}$. The top Lyapunov exponent is $\lambda(r) = \lim_{t \to \infty} t^{-1} \log \|\Phi_t\|$; the limit exists almost surely and is not random (Proposition 2.1).

Theorem 1.1. *Let*

$$A_0 = \begin{pmatrix} -3/4 & -1 \\ 1 & -3/4 \end{pmatrix}, \qquad A_1 = \begin{pmatrix} 5/4 & 0 \\ 0 & -11/4 \end{pmatrix}. \tag{2}$$

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Then λ is continuous on $(0,\infty)$, and for every $X_0 \neq 0$ and every $r > 0$, $t^{-1}\log|X_t| \to \lambda(r)$ almost surely. Moreover

$$\begin{aligned} \lambda(r) &> \tfrac{9}{50} && \text{for } 0 < r \le \tfrac{1}{10}, \\ -0.0756 \le \lambda(r) &\le -0.0055 && \text{for } 4.997 \le r \le 5.003, \\ 0.0361 \le \lambda(r) &\le 0.1192 && \text{for } 19.95 \le r \le 20.05. \end{aligned}$$

Consequently the set $U = \{r > 0 : \lambda(r) > 0\}$ of unstable rates is open and has at least two connected components, one containing $(0, 1/10]$ and another containing $[19.95, 20.05]$, and they are separated by the stable interval $[4.997, 5.003]$.

The matrix A_0 has eigenvalues $-\frac{3}{4} \pm i$ and is Hurwitz; A_1 has eigenvalues $\frac{5}{4}$ and $-\frac{11}{4}$ and is not. Two limits of λ are positive. As $r \to 0$, $\lambda(r) \to \frac{1}{2}\big(-\frac{3}{4} + \frac{5}{4}\big) = \frac{1}{4}$ (Proposition 4.1), and as $r \to \infty$, $\lambda(r)$ tends to the largest eigenvalue $-\frac{3}{4} + \frac{\sqrt{3}}{2}$ of $\bar{A} = (A_0 + A_1)/2$ [4, Proposition 6.4]. The set U therefore has at least two components as soon as $\lambda(r) < 0$ at one rate. Neither limit gives the sign of λ at a rate of order one, and the direct sums of [9], which give k components in dimension $2k$, need dimension at least four.

The sign at $r = 5$ comes from the following estimate. Write $X_t = |X_t|(\cos\Theta_t, \sin\Theta_t)$, so that $\frac{d}{dt}\log|X_t| = h_{I_t}(\Theta_t)$ and $\dot{\Theta}_t = v_{I_t}(\Theta_t)$ with $h_0 = -\frac{3}{4}$, $h_1(\theta) = -\frac{3}{4} + 2\cos 2\theta$, $v_0 = 1$ and $v_1(\theta) = -2\sin 2\theta$; the pair (Θ, I) is the *angular process*. For every $r > 0$ and every pair $f = (f_0, f_1)$ of C^1 functions of period π,

$$\min_{i,\theta} g_i(\theta) \;\le\; \lambda(r) \;\le\; \max_{i,\theta} g_i(\theta), \qquad g_i = h_i + v_i f_i' + r(f_{1-i} - f_i)$$

(Lemma 3.1). A pair f with $g_0 = g_1 = \lambda(r)$ would solve the Poisson equation of the angular process; a pair with rational coefficients for which these bounds are computed exactly is called a *certificate* at rate r. If f_0 and f_1 are trigonometric polynomials with rational coefficients, then so are g_0 and g_1, their coefficients are computed exactly, and g_i differs from its constant coefficient by at most the sum of the absolute values of the others. With f of degree $2K$ this gives $-0.0714 \le \lambda(5) \le -0.0097$ for $K = 128$ and $0.0703 \le \lambda(20) \le 0.0850$ for $K = 32$ (Proposition 3.2); the second bound places an explicit rate in the component of U that contains all large rates. Since g_i depends on r only through the term $r(f_{1-i} - f_i)$, the same two pairs give the signs on $[4.997, 5.003]$ and $[19.95, 20.05]$ (Corollary 3.3). Malrieu defines the unstable rates as those for which $|X_t| \to \infty$ almost surely; this set contains $(0, \frac{1}{10}]$ and $[19.95, 20.05]$ and misses $[4.997, 5.003]$, so it is not an interval either.

Section 2 sets up the angular process and proves that λ exists and is continuous. Section 3 proves the certificate lemma and the signs near $r = 5$ and $r = 20$, and Section 4 the sign at small rates together with the proof of Theorem 1.1. Section 5 computes the whole curve from the stationary density, and Section 6 proves that a Hurwitz pair with two unstable components must contain a non-normal matrix.

Related work. Lawley, Mattingly and Reed [9] prove that switching between Hurwitz matrices is stable at slow and fast rates when the averaged matrix is Hurwitz. In their Example 3.3 they take direct sums of planar blocks to build pairs of $2k \times 2k$ matrices whose unstable set has k separated components. The smallest case with two components is in dimension four. Lagasquie [8] studies a planar pair of the same kind, two Hurwitz matrices all of whose convex combinations are Hurwitz, and observes numerically a second region of instability after replacing the exponential holding times by Gamma holding times; his model keeps two planar matrices but changes the switching law. Benaïm, Le Borgne, Malrieu and Zitt [1] take two planar Hurwitz matrices with a convex combination that has a positive eigenvalue and show that

$|X_t|$ tends to 0 or to infinity according to the rate. Porfiri, Jeter and Belykh find disconnected windows of stability for jump linear systems in discrete time, where an independent matrix is held for an integer number of steps [14], and antiresonance in a nonlinear planar system that switches between unstable modes [15]. Neither setting is the two-state exponential switching of (1). Benaïm and Strickler [3] relate the top exponent of switched linear systems to invariant measures of the angular process and treat fast switching; Chitour, Mazanti, Monmarché and Sigalotti [4] compare deterministic and probabilistic exponents and give the fast-switching limit used in Section 5. Slow and fast limits for cooperative systems are in [2] and [12], and a three-dimensional cooperative example with a stable convex hull and unstable switching is in [13]. None of these gives two planar matrices whose unstable set has two components under the switching law (1).

2. The angular process and the exponent

For $\theta \in \mathbb{R}/\pi\mathbb{Z}$ put $u_\theta = (\cos\theta, \sin\theta)$ and $u_\theta^\perp = (-\sin\theta, \cos\theta)$. For a matrix A let $h_A(\theta) = u_\theta^{\mathsf T} A u_\theta$ and $v_A(\theta) = (u_\theta^\perp)^{\mathsf T} A u_\theta$. Writing $X_t = R_t u_{\Theta_t}$ with $R_t = |X_t|$, the equation $\dot X = A_I X$ becomes

$$\frac{d}{dt}\log R_t = h_{I_t}(\Theta_t), \qquad \dot\Theta_t = v_{I_t}(\Theta_t), \tag{3}$$

with $h_i = h_{A_i}$ and $v_i = v_{A_i}$. The angular process (Θ, I) is a Markov process on $(\mathbb{R}/\pi\mathbb{Z}) \times \{0,1\}$ whose generator acts on pairs $f = (f_0, f_1)$ of C^1 functions with period π by

$$(\mathcal{L}_r f)_i = v_i f_i' + r(f_{1-i} - f_i). \tag{4}$$

For the matrices (2),

$$v_0 = 1, \qquad v_1(\theta) = -2\sin 2\theta, \qquad h_0 = -\tfrac{3}{4}, \qquad h_1(\theta) = -\tfrac{3}{4} + 2\cos 2\theta. \tag{5}$$

In state 0 the direction rotates at unit speed and the norm decays at rate $3/4$; in state 1 the direction moves toward the horizontal axis.

Proposition 2.1. *For the matrices* (2) *and every* $r > 0$ *there is* $\lambda(r) \in \mathbb{R}$ *such that* $t^{-1}\log|X_t| \to \lambda(r)$ *almost surely for every* $X_0 \neq 0$ *and every initial state* I_0*;* $t^{-1}\log\|\Phi_t\| \to \lambda(r)$ *almost surely, and* λ *is continuous on* $(0,\infty)$*.*

Proof. Start the clock at a jump into state 0 and call a *cycle* a holding period T_0 in state 0 followed by a holding period T_1 in state 1; the holding times are independent and exponential with mean $1/r$. Let $\Theta^{(n)}$ be the direction at the start of the n-th period in state 1. During a period in state 0 the direction moves by T_0 modulo π, and T_0 modulo π has density

$$k_r(s) = \frac{r e^{-rs}}{1 - e^{-\pi r}}, \qquad 0 \le s < \pi, \tag{6}$$

so $m_r \le k_r \le M_r$ with $m_r = r/(e^{\pi r} - 1)$ and $M_r = r/(1 - e^{-\pi r})$. Since the period in state 1 is independent of the next period in state 0, the transition kernel of the chain $(\Theta^{(n)})$ is bounded below by $m_r\, d\theta$, uniformly in the starting direction. By Doeblin's condition the chain has a unique stationary law, converges to it in total variation at a geometric rate uniform in the starting point, and its ergodic averages converge almost surely from every starting point [11, Theorem 16.0.2].

The increment of $\log R$ over a cycle is a function of the direction at its start and of (T_0, T_1), bounded in absolute value by $\|A_0\|T_0 + \|A_1\|T_1$, which has finite second moment. The ergodic theorem for the Markov chain formed by the directions and the holding times, together with the strong law for $\sum(T_0 + T_1)$, gives $t^{-1}\log R_t \to \lambda(r)$ along cycle ends, with $\lambda(r)$ equal to the stationary mean of the cycle increment divided by the mean cycle length $2/r$. With

$C = \max_i \|A_i\|$, between cycle ends $\log R$ moves by at most $C(T_0 + T_1)$ for the current cycle, and $\max_{n \le N}(T_0^{(n)} + T_1^{(n)})/N \to 0$ almost surely, so the limit holds along all t. It does not depend on $X_0 \neq 0$; applying it to two independent initial vectors identifies $\lambda(r)$ with the top exponent of the cocycle.

For continuity, fix a compact interval $J \subset (0, \infty)$. The constant m_r is bounded below on J, so the Doeblin bound is uniform there. The holding times have joint density $r^2 e^{-r(t_0+t_1)}$, continuous in r in L^1, while the deterministic maps from the holding times to the next direction and to the cycle increment do not depend on r. Hence the transition kernel is continuous in r in total variation, uniformly in the starting direction, and by the uniform geometric contraction so is the stationary law. The conditional mean of the cycle increment given the starting direction is bounded and continuous in r uniformly in the direction, by dominated convergence with the bound $C(t_0 + t_1)$. Dividing the stationary mean by $2/r$ gives continuity of λ on J. □

3. Certificates

Lemma 3.1. *Let $r > 0$ and let $f = (f_0, f_1)$ be a pair of C^1 functions of period π. Put $g_i = h_i + (\mathcal{L}_r f)_i$. If $\ell \le g_i(\theta) \le u$ for $i = 0, 1$ and all θ, then $\ell \le \lambda(r) \le u$.*

Proof. By Dynkin's formula, $M_t = f_{I_t}(\Theta_t) - f_{I_0}(\Theta_0) - \int_0^t (\mathcal{L}_r f)_{I_s}(\Theta_s)\, ds$ is a martingale. Its jumps occur at the jumps of I, at rate r, and are bounded by $2\|f\|_\infty$, so $\mathbb{E}M_t^2 \le 4r\|f\|_\infty^2 t$ and $M_t/t \to 0$ almost surely by the strong law for square-integrable martingales [5, Theorem 2.18] applied along integer times, together with the bound on the jumps in between. By (3),

$$\frac{1}{t}\log\frac{R_t}{R_0} = \frac{1}{t}\int_0^t g_{I_s}(\Theta_s)\, ds - \frac{f_{I_t}(\Theta_t) - f_{I_0}(\Theta_0) - M_t}{t},$$

and the last term tends to 0. The integral average lies in $[\ell, u]$, and the left side tends to $\lambda(r)$ by Proposition 2.1. □

Take $f_i(\theta) = c_{i,0} + \sum_{k=1}^{K}\big(c_{i,k}\cos 2k\theta + s_{i,k}\sin 2k\theta\big)$ with rational coefficients. Since $v_0 = 1$ and $v_1 = -2\sin 2\theta$, each g_i is a trigonometric polynomial of degree at most $2(K+1)$ with rational coefficients,

$$g_i(\theta) = C_i + \sum_{k=1}^{K+1}\big(\alpha_{i,k}\cos 2k\theta + \beta_{i,k}\sin 2k\theta\big), \tag{7}$$

and therefore

$$C_i - \sum_{k=1}^{K+1}\big(|\alpha_{i,k}| + |\beta_{i,k}|\big) \le g_i \le C_i + \sum_{k=1}^{K+1}\big(|\alpha_{i,k}| + |\beta_{i,k}|\big). \tag{8}$$

Proposition 3.2. *There are rational pairs $f^{(5)}$ with $K = 128$ and $f^{(20)}$ with $K = 32$, all coefficients with denominators dividing 10^{12}, for which the bounds (8) at $r = 5$ and $r = 20$ are the numbers in Table 1. Consequently*

$$-0.071355633951 \le \lambda(5) \le -0.009745897257, \qquad 0.070309824390 \le \lambda(20) \le 0.084999667350.$$

Proof of Proposition 3.2. The coefficients are listed in the repository, in the file `data/poisson_certificates.json`. They are twelve-place decimal roundings of a floating-point solution of the Poisson equation restricted to the harmonics $k \le K$. Given the coefficients, (7) is computed by exact rational arithmetic, using $\frac{d}{d\theta}\cos 2k\theta = -2k\sin 2k\theta$, $\frac{d}{d\theta}\sin 2k\theta = 2k\cos 2k\theta$ and the product formulas for $\sin 2\theta\cos 2k\theta$ and $\sin 2\theta\sin 2k\theta$. The file `code/verify_certificate.py` carries out this computation with exact rationals in the basis $\cos 2k\theta, \sin 2k\theta$, and `code/`

r	state i	lower bound in (8)	upper bound in (8)
5	0	−0.040550774017	−0.040550757183
5	1	−0.071355633951	−0.009745897257
20	0	0.077654745278	0.077654746482
20	1	0.070309824390	0.084999667350

TABLE 1. The bounds (8) for $g_i = h_i + (\mathcal{L}_r f)_i$, with $f = f^{(5)}$ at $r = 5$ and $f = f^{(20)}$ at $r = 20$. Every entry is a rational number with denominator dividing 10^{12}, written in full.

`independent_certificates.py` in the basis z^k, $z = e^{2i\theta}$, where $v_1 = i(z - z^{-1})$; both give the entries of Table 1. Lemma 3.1 with $\ell = \min_i$ and $u = \max_i$ of the table entries gives the bounds. □

Corollary 3.3. $-0.0756 \le \lambda(r) \le -0.0055$ *for* $4.997 \le r \le 5.003$ *and* $0.0361 \le \lambda(r) \le 0.1192$ *for* $19.95 \le r \le 20.05$.

Proof. By (4), $(\mathcal{L}_r f)_i - (\mathcal{L}_{r_0} f)_i = (r - r_0)(f_{1-i} - f_i)$, so with the same f the bounds (8) for g_i at rate r are those at r_0 widened by $|r - r_0|\,\|f_1 - f_0\|_\infty$ on each side. Bounding $\|f_1 - f_0\|_\infty$ by the sum of the absolute values of the Fourier coefficients of $f_1 - f_0$ gives at most 1.3918 for $f^{(5)}$ and at most 0.6824 for $f^{(20)}$ (the exact rationals are in `data/certified_neighborhoods.json`). For $f^{(5)}$ the widened bounds for g_0 and g_1 lie in

$$[-0.071355633951 - 0.003 \cdot 1.3918,\ -0.009745897257 + 0.003 \cdot 1.3918] \subset [-0.0756,\ -0.0055],$$

and for $f^{(20)}$ they lie in

$$[0.070309824390 - 0.05 \cdot 0.6824,\ 0.084999667350 + 0.05 \cdot 0.6824] \subset [0.0361,\ 0.1192],$$

since by Table 1 the bounds for g_0 lie inside those for g_1 at both rates. Lemma 3.1 gives the bounds. □

4. Small rates

Proposition 4.1. *For every* $r > 0$,

$$\lambda(r) \ge \frac{1}{4} - \frac{\pi \log 2}{2} \frac{r^2}{1 - e^{-\pi r}}. \tag{9}$$

In particular $\lambda(r) > 9/50$ *for* $0 < r \le 1/10$, *and* $\lambda(r) \to 1/4$ *as* $r \to 0$.

Proof. Use the cycles of the proof of Proposition 2.1. In state 0, $|e^{A_0 t} x| = e^{-3t/4}|x|$. In state 1, for a unit vector u_θ,

$$\log |e^{A_1 T_1} u_\theta| \ge \log\bigl(e^{5T_1/4} |\cos\theta|\bigr) = \tfrac{5}{4} T_1 + \log|\cos\theta|.$$

The direction $\Theta^{(n)}$ at the start of a period in state 1 is the previous direction shifted by an independent T_0 modulo π, so its conditional density given the past is at most M_r. Since $\log|\cos\theta| \le 0$ and $\int_0^\pi \log|\cos\theta|\, d\theta = -\pi \log 2$,

$$\mathbb{E}\bigl[\log|\cos\Theta^{(n)}| \,\big|\, \text{past}\bigr] \ge -M_r \pi \log 2.$$

The stationary mean of the cycle increment is therefore at least $-\frac{3}{4r} + \frac{5}{4r} - M_r \pi \log 2$, and dividing by the mean cycle length $2/r$ gives (9). Using $1 - e^{-x} \ge x/(1 + x)$ for $x > 0$,

$$\lambda(r) \ge \frac{1}{4} - \frac{r(1 + \pi r)\log 2}{2},$$

and for $0 < r \le 1/10$, with $\pi < 4$ and $\log 2 < 1$, the right side exceeds $\frac{1}{4} - \frac{1}{20} \cdot \frac{14}{10} = \frac{9}{50}$. For the limit, the cycle increment is at most $-\frac{3}{4}T_0 + \frac{5}{4}T_1$, so $\lambda(r) \le \frac{1}{4}$ for every r. □

Proof of Theorem 1.1. Continuity and the almost sure limit are Proposition 2.1. The bounds are Proposition 4.1 and Corollary 3.3. Since λ is continuous, U is open in $(0, \infty)$ and each of its connected components is an open interval. The points $1/10$ and 20 lie in U, and the point 5 between them does not, so they lie in different components. □

5. Numerical picture

The certificates prove signs at chosen rates; the shape of the whole curve is computed numerically. The stationary law of (Θ, I) has densities p_0, p_1 satisfying $p_0' = r(p_1 - p_0)$ and $(v_1 p_1)' = r(p_0 - p_1)$, with constant positive flux $J = p_0 + v_1 p_1$ and $\int_0^\pi p_0 = \int_0^\pi p_1 = \frac{1}{2}$; then

$$\lambda(r) = -\frac{3}{4} + 2\int_0^\pi \cos(2\theta)\, p_1(\theta)\, d\theta.$$

With $H(\theta) = \theta - \frac{1}{4}\log|\tan\theta|$ (so that $H' = 1/v_0 + 1/v_1$) the solution is explicit on each quarter-circle: for $0 < \theta < \pi/2$,

$$F_-(\theta) = e^{-rH(\theta)} \int_\theta^{\pi/2} e^{rH(s)}\, ds, \qquad p_0 = J(1 + rF_-), \qquad p_1 = \frac{JrF_-}{2\sin 2\theta},$$

and for $\pi/2 < \theta < \pi$,

$$F_+(\theta) = e^{-rH(\theta)} \int_{\pi/2}^\theta e^{rH(s)}\, ds, \qquad p_0 = J(1 - rF_+), \qquad p_1 = -\frac{JrF_+}{2\sin 2\theta},$$

with J fixed by normalization. On the second branch $H' > 1$, which gives $rF_+ < 1$ and $p_0 > 0$. We evaluate these integrals as ordinary differential equations in the coordinate $x = \log|\tan\theta|$ on $(0, \pi/2)$ and $x = -\log|\tan\theta|$ on $(\pi/2, \pi)$, so that $\cos 2\theta = -\tanh x$ on the first branch and $\tanh x$ on the second, which resolves the power singularities at the attracting and repelling directions of A_1 (`code/candidate_density.py`). The same integrals computed in a second coordinate (`code/independent_density.py`) give values of λ that differ from these by less than 10^{-12} at $r = 1/10$, 5 and 20. Table 2 compares the density method with two other estimates at those rates.

r	stationary density	Markov chain, 16384 cells	Monte Carlo (standard error)
0.1	0.216444855966	0.216417929898	0.21600 (0.00036)
5	−0.040611505820	−0.040711536760	−0.04159 (0.00066)
20	0.077660185766	0.077491305516	0.07702 (0.00061)

Table 2. Three estimates of $\lambda(r)$. The Markov chain discretizes the angle into cells with upwind moves at rate $|v_i|/\Delta\theta$; its remaining bias is mesh error. The Monte Carlo runs use exact matrix flows between exponential holding times, two seeds of 2×10^6 cycles each; the standard errors are batch estimates.

On a grid of 403 rates in $[10^{-3}, 10^2]$ the density method gives exactly two sign changes, near $r = 1.2228$ and $r = 7.7616$ (to ten places 1.2228361663 and 7.7615867624; Figure 1). A grid of 151 rates in $[10^{-3}, 10^3]$ gives the same two sign changes. The two crossings and the absence of others are numerical observations. As $r \to 0$, $\lambda(r) \to \frac{1}{2}(-\frac{3}{4} + \frac{5}{4}) = \frac{1}{4}$ by Proposition 4.1. As $r \to \infty$, $\lambda(r)$ tends to the largest eigenvalue $-\frac{3}{4} + \frac{\sqrt{3}}{2} \approx 0.1160$ of $\bar{A}$; this follows from [4,

Proposition 6.4], whose condition (C) holds because the lower eigenspace of $\bar{A}$ is a line and A_0 leaves no real line invariant (their redraw intensity μ is $2r$ in our notation). So both ends of the curve are unstable and the exponent dips below zero in between.

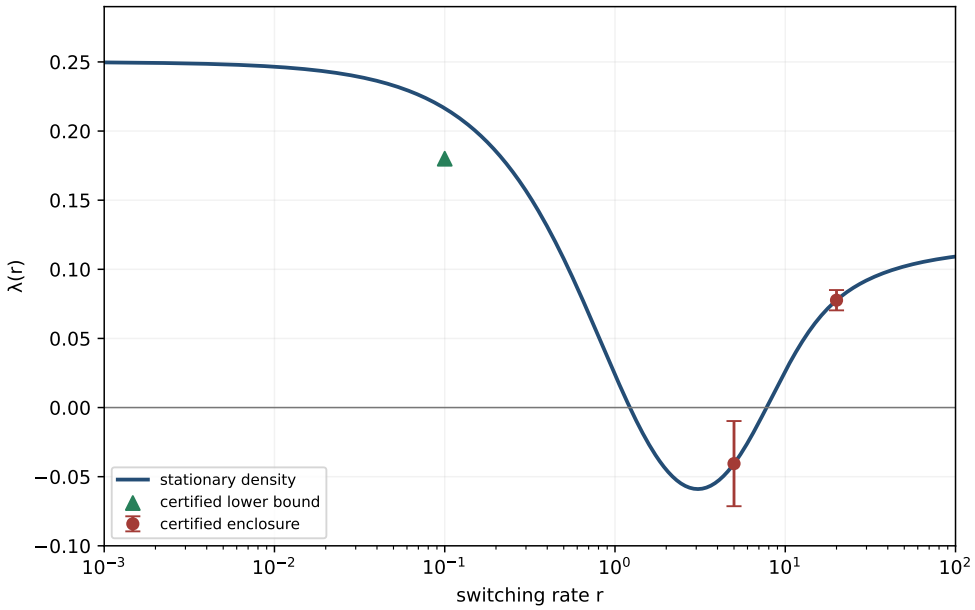


FIGURE 1. The exponent $\lambda(r)$ computed from the stationary density, on a logarithmic rate axis. The marked points are the rates where the sign is proved.

6. The Hurwitz case

In the example of [9] that precedes Open question 3 both matrices are Hurwitz. Whether two Hurwitz matrices can have an unstable set with two components is open.

First, no pair of Hurwitz matrices of the form $A_0 - c_0\mathrm{Id}$, $A_1 - c_1\mathrm{Id}$ is unstable at any rate. The shift lowers $\lambda(r)$ by $(c_0 + c_1)/2$ for every r, because the two states are occupied half of the time each. The shifted matrices are Hurwitz exactly when $c_0 > -\frac{3}{4}$ and $c_1 > \frac{5}{4}$, so $(c_0 + c_1)/2 > \frac{1}{4}$, and the shifted exponent is less than $\lambda(r) - \frac{1}{4} \le 0$ by the bound $\lambda \le \frac{1}{4}$ in the proof of Proposition 4.1. More generally, a normal matrix has symmetric part with eigenvalues equal to the real parts of its own eigenvalues, so a normal Hurwitz matrix A satisfies $x^{\mathsf{T}} A x \le -\epsilon |x|^2$ for some $\epsilon > 0$. If both matrices are normal and Hurwitz, then $\frac{d}{dt}|X_t|^2 \le -2\epsilon |X_t|^2$ along every switching path, and $\lambda(r) \le -\epsilon$ for every r. A Hurwitz pair whose unstable set has two components must therefore contain a non-normal matrix, as the pair of [9] does.

Second, a numerical search gave no example. It starts from 515 pairs of traceless planar matrices, drawn as in `code/search_ctmc.py`: in a third of the draws A_0 generates the rotations, in a third $A_0 = \mathrm{diag}(1, -1)$, and in a third both matrices have integer entries. Each matrix is shifted by a multiple of the identity to spectral abscissa $-\gamma$, $\gamma \in \{0.01, 0.05, 0.1\}$, which lowers the curve λ by a constant. At 49 rates in $[10^{-3}, 10^{3}]$, with a 256-cell discretization of the direction, 1350 of the 1545 shifted pairs have no unstable rate, 195 have one run of unstable rates, and none has two (`data/hurwitz_search.json`). Components narrower than the spacing of the rates are not resolved.

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Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies

The paper comes out of Math Brain, an ongoing project on the use of AI systems for research mathematics, and its results were developed with the Brain [19] and Math Brain [17]. A Codex agent (OpenAI) found the pair (2), constructed the certificates and proved Theorem 1.1; a second Codex agent checked the proofs and recomputed the certificates, the stationary density and the Monte Carlo estimates with its own programs; a Claude agent (Anthropic) computed the stationary density with its own discretization and prepared the manuscript. The author directed the work and is responsible for the content of the paper and for decisions concerning its dissemination.

Code and data availability

The certificates, the programs that check them and the numerical computations are in the repository

`https://github.com/yspennstate/planar-switching-two-instability-intervals`.

The exact checks need only the Python standard library; the interval audit uses `mpmath` [7], and the numerical computations use NumPy [6] and SciPy [16].

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Appendix A. Notation

Symbol	Meaning	First appears
X_t, I_t	the planar component and the switching state	Section 1
A_0, A_1	the two matrices; from Theorem 1.1 on, the pair (2)	Section 1
r	the rate of each jump of I	Section 1
Φ_t, $\lambda(r)$	the fundamental matrix and the top Lyapunov exponent	Section 1
U	the set of rates with $\lambda(r) > 0$	Theorem 1.1
$\bar{A}$	$(A_0 + A_1)/2$	Section 1
Θ_t	the direction of X_t modulo π	Section 1
h_i, v_i	radial growth rate and angular velocity in state i	Section 1
$f = (f_0, f_1)$, g_i	a pair of C^1 functions of period π and $h_i + v_i f_i' + r(f_{1-i} - f_i)$	Section 1
K	the number of harmonics in a certificate	Section 1
u_θ, $u_\theta^\perp$	the unit vector at angle θ and its rotation by $\pi/2$	Section 2
R_t	$\lvert X_t \rvert$	(3)
T_0, T_1	holding times in states 0 and 1 during one cycle	Proposition 2.1
$\Theta^{(n)}$	the direction at the start of the n-th period in state 1	Proposition 2.1
k_r	the density of T_0 modulo π	(6)
m_r, M_r	lower and upper bounds of the density (6)	Proposition 2.1
$\mathcal{L}_r$	generator of the angular process (Θ, I)	(4)
C_i, $\alpha_{i,k}$, $\beta_{i,k}$	Fourier coefficients of g_i	(7)
$f^{(5)}$, $f^{(20)}$	the certificates at $r = 5$ and $r = 20$	Proposition 3.2
p_0, p_1, J	stationary densities of (Θ, I) and their flux	Section 5
H, $F_\pm$	the functions in the explicit stationary density	Section 5

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